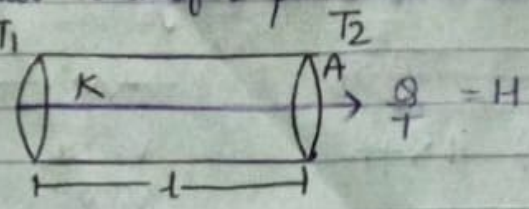
Conduction :- Heat transfer per unit time :-(higher to lower)



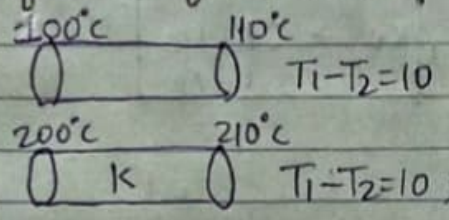
$$H \Rightarrow \frac{Q}{T} \Rightarrow \frac{KA}{l}(T_1 - T_2)$$

unit $\Rightarrow \frac{J}{s}$ = Watt.

- Thermal conductivity is depend on material, T.C is high of Metal

Q. The two ends of a metal rod are maintained at 100°C & 110°C. The rate of heat flows in the rod is found to be 4.0 J/s. If the ends are maintained at 2000°C & 2100°C, the ratio of heat flow will be :-

A:-

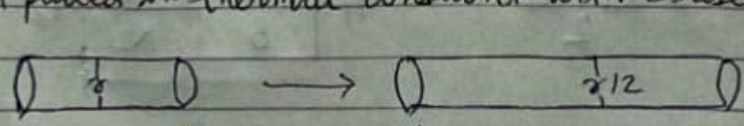


$$\frac{Q}{t} = \frac{KA}{l}(T_1 - T_2)$$

4.0 J/s Ans

Q. A cylindrical Metallic rod in thermal contact with two reservoirs of heat at its two ends conducts an amount of heat Q in time t. The metallic rod is melted & the material is formed into rod of half the radius of original rod. What is amount of heat conducted by new rod when placed in thermal constant with 2 reservoirs in time t?

A:-



$\{A = \pi r^2\}$

$\{A = \frac{1}{4}\}$

$$V_1 = V_2$$

$$A_1 l_1 = A_2 l_2$$

$$\frac{A}{4} l^4 = \text{const}$$

$$\frac{Q}{t} = \frac{KA}{l}(T_1 - T_2) \Rightarrow \frac{Q}{T} = \frac{KA}{4 \times l \times 4}(T_1 - T_2) \Rightarrow \frac{Q}{T} = \frac{1}{4}$$

Stefan Boltzmann's law:-

$$E = \sigma T^4$$

$$\sigma = 5.67 \times 10^{-8} \text{ (stefan's const)}$$

Emissive power of normal body

$$\frac{E}{E_b} = e \quad \{\text{black body}\}$$

$$E = eE_b = e\sigma T^4$$

Q. A black body, at a temp of 227°c radiates heat at rate of 7 cal/cm²/s. At temp of 727°c, the rate of heat is radiated in the same units will be:-

A:-

$$E_b \propto T^4$$

$$E_b = \sigma T^4$$

$$E_b \propto (2T)^4$$

$$E_b = 16 \text{ times}$$

Given, $\sigma = 7 \text{ cal/cm}^2/\text{s}$

$$E_b = 7 \times 16$$

$$E_b = 112$$

Q. Total radiant energy per unit area, per unit time normal to direction of incidence, recieved at distance R from centre of a star of radius r, whose outer surface radiates as black body at temp T Kelvin is :-

A:- solar intensity at r distance $= I = \frac{\text{Energy}}{T \times \text{Area}} = \frac{\text{Power}}{\text{Area}}$

$$\text{Area of sphere/star} = \frac{P}{4\pi R^2}$$

$$\frac{\sigma T^4 \, 4\pi r^2}{4\pi R^2} = \frac{\sigma r^2 T^4 A_r}{R^2}$$

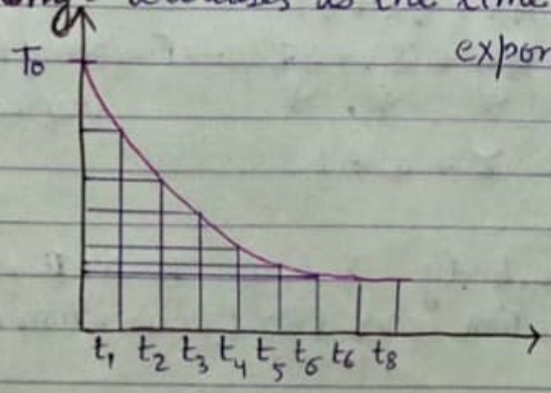
Heat Transfer

~GG sir.

Newton's law of cooling :- for small temp. difference b/w a body & surrounding.

Rate of cooling :- Decreases as the time increases

exponential graph



Q. An object kept in large room having, $T = 25°C$ takes 12 min to cool from 80°C to 70°C, The time taken to cool for the same object from 70°C to 60°C would be nearly

A. Given, $T = 25°C$. (Temp of surrounding)

$80°C \xrightarrow[12\,min]{} 70°C \xrightarrow[\text{more than } 12\,min]{} 60°C$ {Temp ↓, Time ↑}

Correct answer by – Newton's law of Cooling.

$$\frac{\Delta T}{\Delta t} \propto (T - T_s)$$

$$\frac{T_1 - T_2}{t} = K\left(T_{av} - T_s\right)$$

$$\frac{T_1 - T_2}{t} = K\left(\frac{T_1 + T_2}{2} - T_s\right)$$

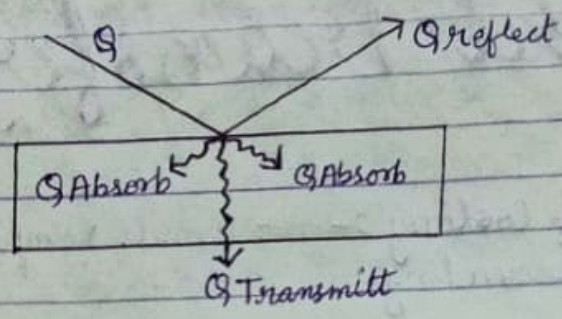
$$\frac{80-70}{12} = K\left(\frac{80+70}{2} - 25°C\right) \Rightarrow \frac{70-60}{t} = K\left(\frac{70+60}{2} - 25°C\right)$$

$$\frac{10}{12} = K(75-25) \quad \text{—(i)} \Rightarrow \frac{10}{t} = K(65-25) \quad \text{—(ii)}$$

Dividing (i) & (ii) $\rightarrow \frac{10}{12} \times \frac{t}{10} = \frac{50}{40} \Rightarrow \frac{t}{12} = \frac{5}{4} \Rightarrow t = \frac{5 \times 12^{3}}{4} = 15$

$t = 15$ min Ans

Radiation:-



$$Q = Q_r + Q_a + Q_T$$

$$\frac{Q}{Q} = \frac{Q_r}{Q} + \frac{Q_a}{Q} + \frac{Q_T}{Q}$$ → Fraction of Heat reflected reflectivity

$$1 = r + a + t$$

NOTE:- $t = 0$ for opaque body & here $1 = r + a$, coz only Absorption & reflection occur (no transmittion occur)

Black body:- No reflection occur, only absorption occur & due to opaque object (no transmission occur)

hence, $r + a = 1$

$r = 0$

$a = 1$.

NOTE:- Snow is Black body object as light is 99.9% absorbed

$r = 0, t = 0, a = 1$ → Perfect Black body

$r = 1, t = 0, a = 0$ → Perfect reflector

$r = 0, t = 1, a = 0$ → Perfect transmittor.

Emissive Power:- Amount of heat emmitted by body per unit area per unit time. is called Emissive power of Body.

$$E = \frac{Q}{At}$$

Unit:- $\frac{J}{M^2 s} = \frac{W}{M^2}$ $\left\{ W = \frac{J}{s} \right\}$

Body which absorbed more can emmitt more at same temp.

$$\frac{E_n}{E_b} = \text{emissivity } (e)$$

where, n = normal body, b = black body